

## H1 Fundamental theorem of calculus

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Time: **401 minutes**

Marks: **339 marks**

Comments:

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**Q1.**

The gradient,  $\frac{dy}{dx}$ , of a curve at the point  $(x, y)$  is given by

$$\frac{dy}{dx} = 10x^4 - 6x^2 + 5$$

The curve passes through the point  $P(1, 4)$ .

- (a) Find the equation of the tangent to the curve at the point  $P$ , giving your answer in the form  $y = mx + c$ .

(3)

- (b) Find the equation of the curve.

(5)

(Total 8 marks)

**Q2.**

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_1^5 \frac{1}{x^2 + 1} dx$$

giving your answer to three significant figures.

(4)

- (b) (i) Find  $\int \left( x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$ , giving the coefficient of each term in its simplest form.

(3)

- (ii) Hence find the value of  $\int_1^4 \left( x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$ .

(2)

(Total 9 marks)

**Q3.**

- (a) (i) Using the binomial expansion, or otherwise, express  $(2 + y)^3$  in the form  $a + by + cy^2 + y^3$ , where  $a$ ,  $b$  and  $c$  are integers.

(2)

- (ii) Hence show that  $(2 + x^{-2})^3 + (2 - x^{-2})^3$  can be expressed in the form  $p + qx^{-4}$ , where  $p$  and  $q$  are integers.

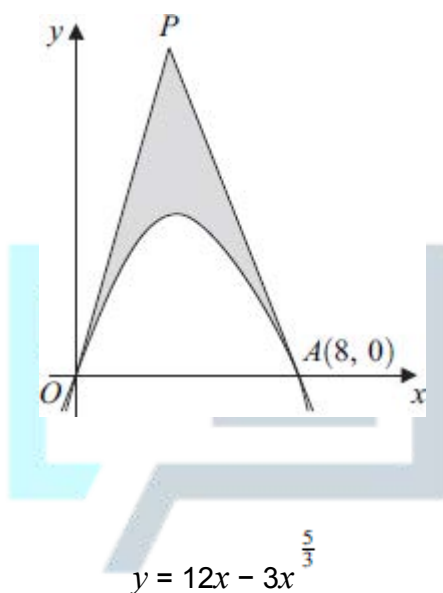
(3)

- (b) (i) Hence find  $\int \left[ (2 + x^{-2})^3 + (2 - x^{-2})^3 \right] dx$ . (2)

- (ii) Hence find the value of  $\int_1^2 \left[ (2 + x^{-2})^3 + (2 - x^{-2})^3 \right] dx$ . (2)
- (Total 9 marks)**

**Q4.**

The diagram shows part of a curve crossing the  $x$ -axis at the origin  $O$  and at the point  $A(8, 0)$ . Tangents to the curve at  $O$  and  $A$  meet at the point  $P$ , as shown in the diagram.



The curve has equation

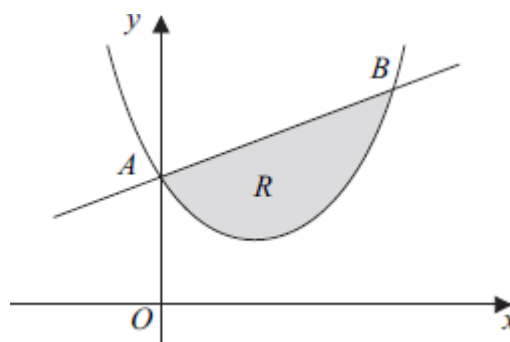
$$y = 12x - 3x^{\frac{5}{3}}$$

- (a) Find  $\frac{dy}{dx}$ . (2)
- (b) (i) Find the value of  $\frac{dy}{dx}$  at the point  $O$  and hence write down an equation of the tangent at  $O$ . (2)
- (ii) Show that the equation of the tangent at  $A(8, 0)$  is  $y + 8x = 64$ . (3)
- (c) Find  $\int \left( 12x - 3x^{\frac{5}{3}} \right) dx$ . (3)
- (d) Calculate the area of the shaded region bounded by the curve from  $O$  to  $A$  and the tangents  $OP$  and  $AP$ .

(7)  
(Total 17 marks)

**Q5.**

- (a) (i) Express  $x^2 - 3x + 5$  in the form  $(x - p)^2 + q$ . (2)
- (ii) Hence write down the equation of the line of symmetry of the curve with equation  $y = x^2 - 3x + 5$ . (1)
- (b) The curve  $C$  with equation  $y = x^2 - 3x + 5$  and the straight line  $y = x + 5$  intersect at the point  $A(0, 5)$  and at the point  $B$ , as shown in the diagram below.



- (i) Find the coordinates of the point  $B$ . (3)
- (ii) Find  $\int (x^2 - 3x + 5) dx$ . (3)
- (iii) Find the area of the shaded region  $R$  bounded by the curve  $C$  and the line segment  $AB$ . (4)

(4)  
(Total 13 marks)

**Q6.**

- (a) Expand  $\left(x^{\frac{3}{2}} - 1\right)^2$ . (2)
- (b) Hence find  $\int \left(x^{\frac{3}{2}} - 1\right)^2 dx$ . (3)

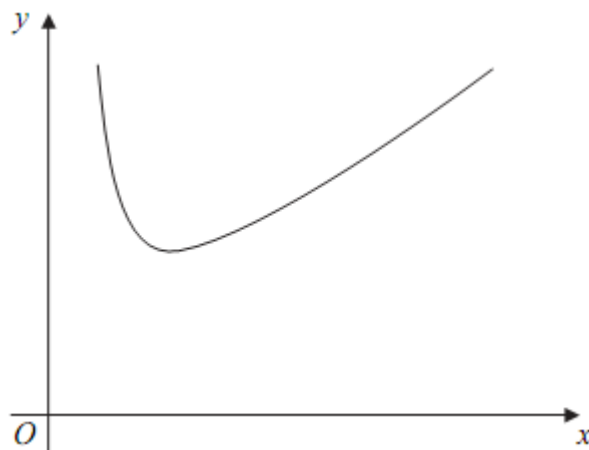
- (c) Hence find the value of  $\int_1^4 \left(x^{\frac{3}{2}} - 1\right)^2 dx$ .

(2)

(Total 7 marks)

**Q7.**

A curve  $C$  is defined for  $x > 0$  by the equation  $y = x + 3 + \frac{8}{x^4}$  and is sketched below.



- (a) Given that  $y = x + 3 + \frac{8}{x^4}$ , find  $\frac{dy}{dx}$ . (3)

- (b) Find an equation of the tangent at the point on the curve  $C$  where  $x = 1$ . (3)

- (c) The curve  $C$  has a minimum point  $M$ . Find the coordinates of  $M$ . (4)

- (d) (i) Find  $\int \left(x + 3 + \frac{8}{x^4}\right) dx$ . (3)

- (ii) Hence find the area of the region bounded by the curve  $C$ , the  $x$ -axis and the lines  $x = 1$  and  $x = 2$ . (2)

- (e) The curve  $C$  is translated by  $\begin{bmatrix} 0 \\ k \end{bmatrix}$  to give the curve  $y = f(x)$ . Given that the  $x$ -axis is a tangent to the curve  $y = f(x)$ , state the value of the constant  $k$ . (1)

(Total 16 marks)

**Q8.**

- (a) The expression  $(2 + x^2)^3$  can be written in the form

$$8 + px^2 + qx^4 + x^6$$

Show that  $p = 12$  and find the value of the integer  $q$ .

(3)

- (b) (i) Hence find  $\int \frac{(2 + x^2)^3}{x^4} dx$ .

(5)

- (ii) Hence find the exact value of  $\int_1^2 \frac{(2 + x^2)^3}{x^4} dx$ .

(2)

(Total 10 marks)

**Q9.**

- (a) The expression  $\left(1 - \frac{1}{x^2}\right)^3$  can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} - \frac{1}{x^6}$$

Find the values of the integers  $p$  and  $q$ .

(2)

- (b) (i) Hence find  $\int \left(1 - \frac{1}{x^2}\right)^3 dx$ .

(4)

- (ii) Hence find the value of  $\int_{\frac{1}{2}}^1 \left(1 - \frac{1}{x^2}\right)^3 dx$ .

(2)

(Total 8 marks)

**Q10.**

- (a) The polynomial  $p(x)$  is given by  $p(x) = x^3 + x - 10$ .

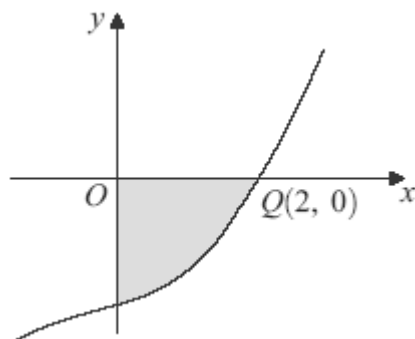
- (i) Use the Factor Theorem to show that  $x - 2$  is a factor of  $p(x)$ .

(2)

(ii) Express  $p(x)$  in the form  $(x - 2)(x^2 + ax + b)$ , where  $a$  and  $b$  are constants.

(2)

- (b) The curve  $C$  with equation  $y = x^3 + x - 10$ , sketched below, crosses the  $x$ -axis at the point  $Q(2, 0)$ .



(i) Find the gradient of the curve  $C$  at the point  $Q$ .

(4)

(ii) Hence find an equation of the tangent to the curve  $C$  at the point  $Q$ .

(2)

(iii) Find  $\int (x^3 + x - 10) dx$ .

(3)

(iv) Hence find the area of the shaded region bounded by the curve  $C$  and the coordinate axes.

(2)

(Total 15 marks)

**Q11.**

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(a) Write down the value of  $n$  given that  $\frac{1}{x^4} = x^n$ .

(1)

(b) Expand  $\left(1 + \frac{3}{x^2}\right)^2$ .

(2)

(c) Hence find  $\int \left(1 + \frac{3}{x^2}\right)^2 dx$ .

(3)

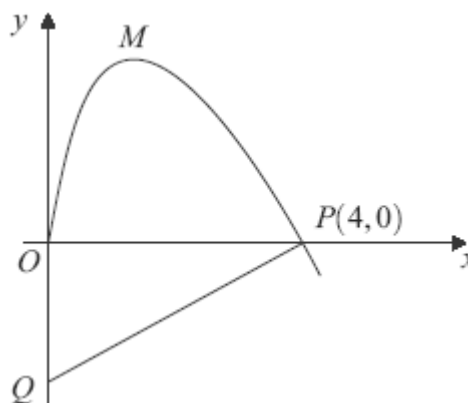
(d) Hence find the exact value of  $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx$ .

**(2)**  
**(Total 8 marks)**

**Q12.**

A curve, drawn from the origin  $O$ , crosses the  $x$ -axis at the point  $P(4, 0)$ .

The normal to the curve at  $P$  meets the  $y$ -axis at the point  $Q$ , as shown in the diagram.



The curve, defined for  $x \geq 0$ , has equation

$$y = 4x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

- (a) (i) Find  $\frac{dy}{dx}$ . (3)
- (ii) Show that the gradient of the curve at  $P(4, 0)$  is  $-2$ . (2)
- (iii) Find an equation of the normal to the curve at  $P(4, 0)$ . (3)
- (iv) Find the  $y$ -coordinate of  $Q$  and hence find the area of triangle  $OPQ$ . (3)
- (v) The curve has a maximum point  $M$ . Find the  $x$ -coordinate of  $M$ . (3)
- (b) (i) Find  $\int \left( 4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx$ . (3)
- (ii) Find the total area of the region bounded by the curve and the lines  $PQ$  and  $QO$ . (3)



(Total 20 marks)

**Q13.**

- (a) The expression  $\left(1 + \frac{4}{x^2}\right)^3$  can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^6}$$

By using the binomial expansion, or otherwise, find the values of the integers  $p$  and  $q$ .

(3)

- (b) (i) Hence find  $\int \left(1 + \frac{4}{x^2}\right)^3 dx$ .

(4)

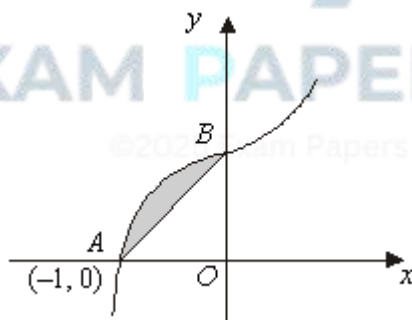
- (ii) Hence find the value of  $\int_1^2 \left(1 + \frac{4}{x^2}\right)^3 dx$ .

(2)

(Total 9 marks)

**Q14.**

The curve with equation  $y = 3x^5 + 2x + 5$  is sketched below.



The curve cuts the  $x$ -axis at the point  $A (-1, 0)$  and cuts the  $y$ -axis at the point  $B$ .

- (a) (i) State the coordinates of the point  $B$  and hence find the area of the triangle  $AOB$ , where  $O$  is the origin.

(3)

- (ii) Find  $\int (3x^5 + 2x + 5) dx$

(3)

- (iii) Hence find the area of the shaded region bounded by the curve and the line  $AB$ .

(4)

- (b) (i) Find the gradient of the curve with equation  $y = 3x^5 + 2x + 5$  at the point  $A(-1, 0)$ .

(3)

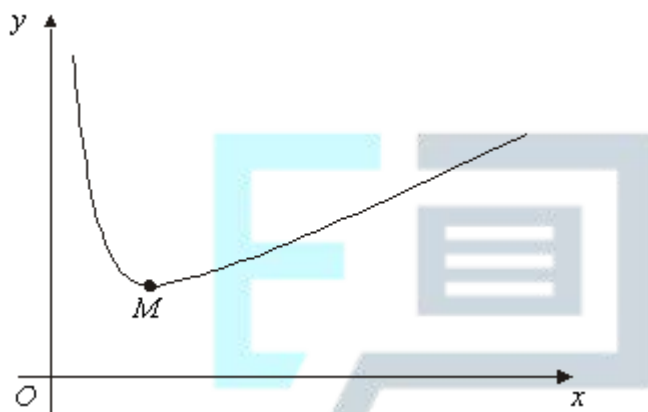
- (ii) Hence find an equation of the tangent to the curve at the point  $A$ .

(1)

(Total 14 marks)

### Q15.

A curve  $C$  is defined for  $x > 0$  by the equation  $y = x + 1 + \frac{4}{x^2}$  and is sketched below.



- (a) (i) Given that  $y = x + 1 + \frac{4}{x^2}$ , find  $\frac{dy}{dx}$ .

(3)

- (ii) The curve  $C$  has a minimum point  $M$ . Find the coordinates of  $M$ .

(4)

- (iii) Find an equation of the normal to  $C$  at the point  $(1, 6)$ .

(4)

- (b) (i) Find  $\int \left( x + 1 + \frac{4}{x^2} \right) dx$ .

(3)

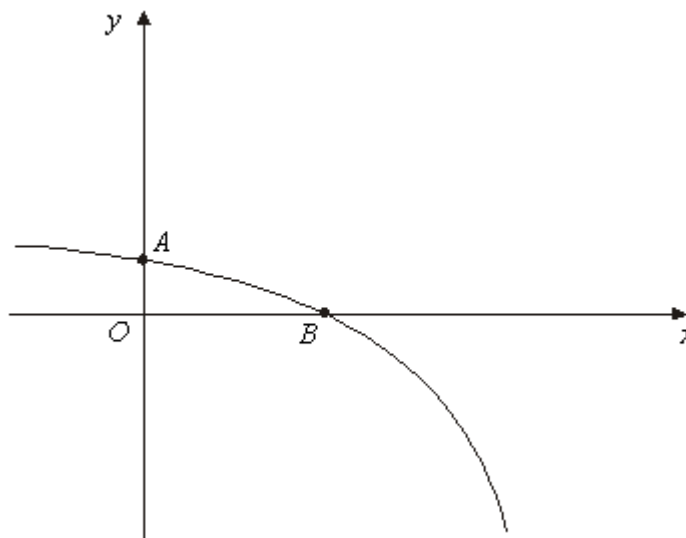
- (ii) Hence find the area of the region bounded by the curve  $C$ , the lines  $x = 1$  and  $x = 4$  and the  $x$ -axis.

(2)

(Total 16 marks)

**Q16.**

The sketch shows the graph of  $y = 4 - e^{2x}$ . The curve crosses the  $y$ -axis at the point  $A$  and the  $x$ -axis at the point  $B$ .



- (a) (i) Find  $\int (4 - e^{2x}) dx$ . (2)
- (ii) Hence show that  $\int_0^{\ln 2} (4 - e^{2x}) dx = 4 \ln 2 - \frac{3}{2}$ . (2)
- (b) (i) Write down the  $y$ -coordinate of  $A$ . (1)
- (ii) Show that  $x = \ln 2$  at  $B$ . (2)
- (c) Find the equation of the normal to the curve  $y = 4 - e^{2x}$  at the point  $B$ . (4)
- (d) Find the area of the region enclosed by the curve  $y = 4 - e^{2x}$ , the normal to the curve at  $B$  and the  $y$ -axis. (3)
- (Total 14 marks)**

**Q17.**

- (a) The polynomial  $f(x)$  is given by  $f(x) = x^3 + 4x - 5$ .
- (i) Use the Factor Theorem to show that  $x - 1$  is a factor of  $f(x)$ . (2)

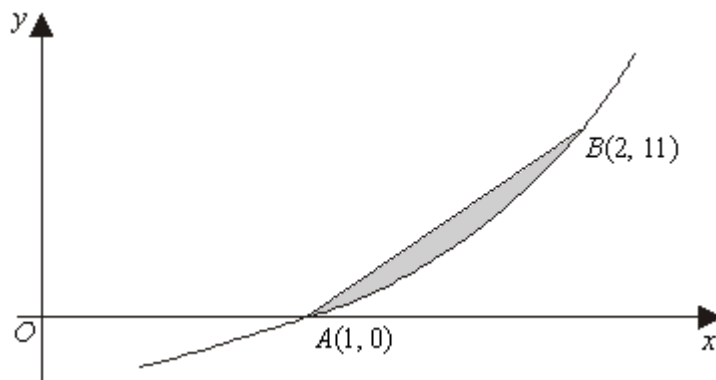
(ii) Express  $f(x)$  in the form  $(x - 1)(x^2 + px + q)$ , where  $p$  and  $q$  are integers.

(2)

(iii) Hence show that the equation  $f(x) = 0$  has exactly one real root and state its value.

(3)

(b) The curve with equation  $y = x^3 + 4x - 5$  is sketched below.



The curve cuts the  $x$ -axis at the point  $A(1, 0)$  and the point  $B(2, 11)$  lies on the curve.

(i) Find  $\int (x^3 + 4x - 5) dx$ .

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line  $AB$ .

(4)

(Total 14 marks)

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**Q18.**

(a) Simplify:

(i)  $x^{\frac{3}{2}} \times x^{\frac{1}{2}};$

(1)

(ii)  $x^{\frac{3}{2}} \div x;$

(1)

(iii)  $\left(x^{\frac{3}{2}}\right)^2$

(1)

(b) (i) Find  $\int 3x^{\frac{1}{2}} dx$ .

(3)

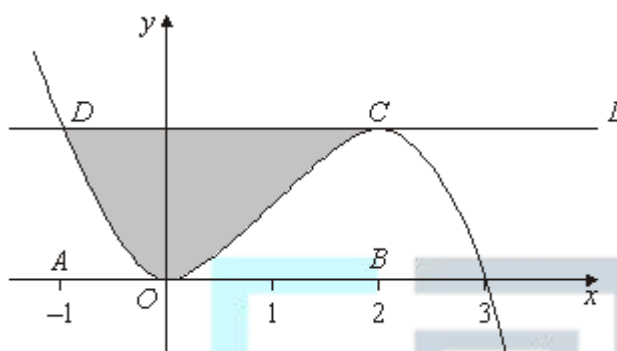
(ii) Hence find the value of  $\int_1^9 3x^{\frac{1}{2}} dx$ .

(2)

(Total 8 marks)

**Q19.**

The diagram shows the curve with equation  $y = 3x^2 - x^3$  and the line  $L$ .



The points  $A$  and  $B$  have coordinates  $(-1, 0)$  and  $(2, 0)$  respectively. The curve touches the  $x$ -axis at the origin  $O$  and crosses the  $x$ -axis at the point  $(3, 0)$ . The line  $L$  cuts the curve at the point  $D$  where  $x = -1$  and touches the curve at  $C$  where  $x = 2$ .

(a) Find the area of the rectangle  $ABCD$ .

(2)

(b) (i) Find  $\int (3x^2 - x^3) dx$ .

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line  $L$ .

(4)

(c) For the curve above with equation  $y = 3x^2 - x^3$ :

(i) find  $\frac{dy}{dx}$ ;

(2)

(ii) hence find an equation of the tangent at the point on the curve where  $x = 1$ ;

(3)

(iii) show that  $y$  is decreasing when  $x^2 - 2x > 0$ .

(2)

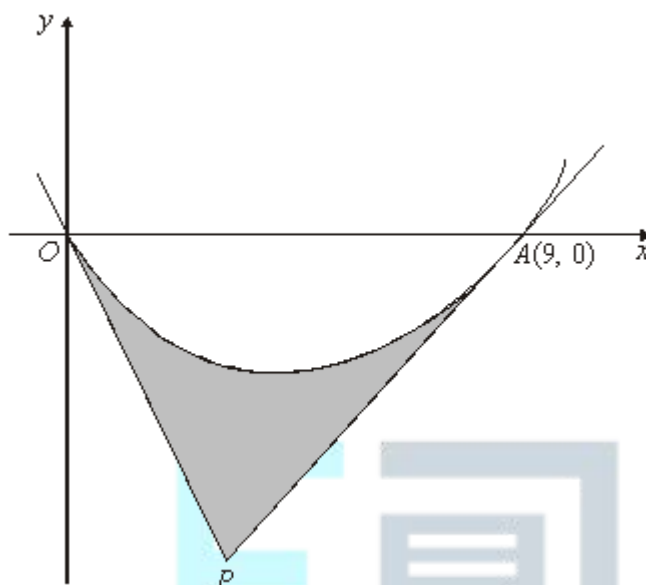
- (d) Solve the inequality  $x^2 - 2x > 0$ .

(2)

(Total 18 marks)

**Q20.**

A curve, drawn from the origin  $O$ , crosses the  $x$ -axis at the point  $A(9, 0)$ . Tangents to the curve at  $O$  and  $A$  meet at the point  $P$ , as shown in the diagram.



The curve, defined for  $x \geq 0$ , has equation

$$y = x^{\frac{3}{2}} - 3x$$

- (a) Find  $\frac{dy}{dx}$ .

(2)

- (b) (i) Find the value of  $\frac{dy}{dx}$  at the point  $O$  and hence write down an equation of the tangent at  $O$ .

(2)

- (ii) Show that the equation of the tangent at  $A(9, 0)$  is  $2y = 3x - 27$ .

(3)

- (iii) Hence find the coordinates of the point  $P$  where the two tangents meet.

(3)

- (c) Find  $\int \left( x^{\frac{3}{2}} - 3x \right) dx$ .

(3)

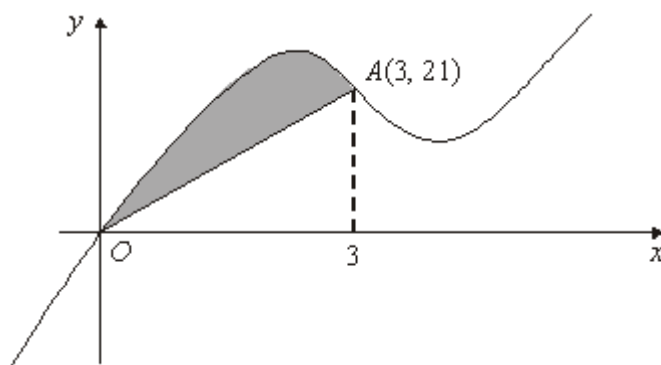
- (d) Calculate the area of the shaded region bounded by the curve and the tangents  $OP$  and  $AP$ .

(5)

(Total 18 marks)

**Q21.**

The curve with equation  $y = x^3 - 10x^2 + 28x$  is sketched below.



The curve crosses the  $x$ -axis at the origin  $O$  and the point  $A(3, 21)$  lies on the curve.

- (a) (i) Find  $\frac{dy}{dx}$ .

(3)

- (ii) Hence show that the curve has a stationary point when  $x = 2$  and find the  $x$ -coordinate of the other stationary point.

(4)

- (b) (i) Find  $\int (x^3 - 10x^2 + 28x) dx$ .

(3)

- (ii) Hence show that  $\int_0^3 (x^3 - 10x^2 + 28x) dx = 56\frac{1}{4}$ .

(2)

- (iii) Hence determine the area of the shaded region bounded by the curve and the line  $OA$ .

(3)

(Total 15 marks)

**Q22.**

- (a) Given that  $y = x \ln x$ , find  $\frac{dy}{dx}$ .

(2)

- (b) Hence, or otherwise, find  $\int_{1}^{\ln} x \, dx$ .

(2)

- (c) Find the exact value of  $\int_{1}^{5} \frac{1}{x} \, dx$ .

(2)

(Total 6 marks)

**Q23.**

- (a) The function  $f$  is defined for all values of  $x$  by  $f(x) = x^3 - 3x^2 - 6x + 8$ .

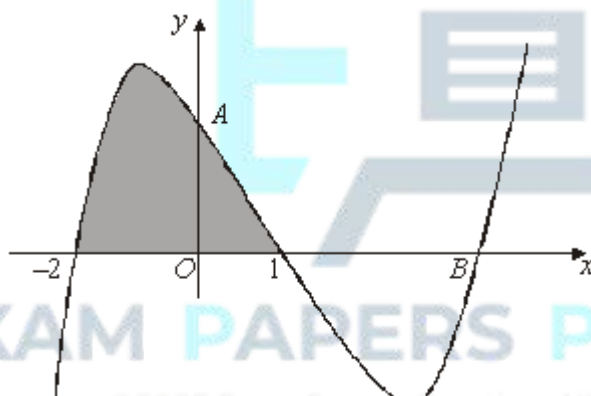
- (i) Given that  $f(1) = 0$  and  $f(-2) = 0$ , write down two linear factors of  $f(x)$ .

(2)

- (ii) Hence express  $x^3 - 3x^2 - 6x + 8$  as the product of three linear factors.

(2)

- (b) The curve with equation  $y = x^3 - 3x^2 - 6x + 8$  is sketched below.



- (i) The curve intersects the  $y$ -axis at the point  $A$ . Find the  $y$ -coordinate of  $A$ .

(1)

- (ii) The curve crosses the  $x$ -axis when  $x = -2$ , when  $x = 1$  and also at the point  $B$ . Use the results from part (a) to find the  $x$ -coordinate of  $B$ .

(1)

- (c) (i) Find  $\int (x^3 - 3x^2 - 6x + 8) \, dx$

(4)

- (ii) Hence find the area of the shaded region bounded by the curve and the  $x$ -axis.

(3)

(Total 13 marks)

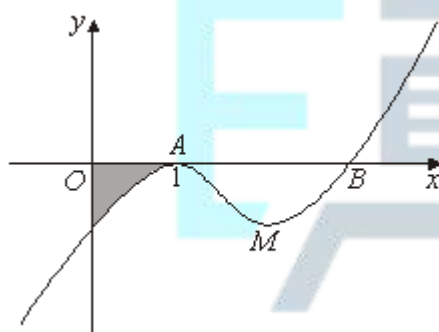


**Q24.**

- (a) Write  $\sqrt{x}$  in the form  $x^k$ , where  $k$  is a fraction. (1)
- (b) Hence express  $\sqrt{x}(x-1)$  in the form  $x^p - x^q$ . (2)
- (c) Find  $\int \sqrt{x}(x-1)dx$ . (3)
- (d) Hence show that  $\int_1^2 \sqrt{x}(x-1)dx = \frac{4}{15}(\sqrt{2}+1)$ . (3)
- (Total 9 marks)**

**Q25.**

The curve with equation  $y = x^3 - 5x^2 + 7x - 3$  is sketched below.



The curve touches the  $x$ -axis at the point  $A$   $(1, 0)$  and cuts the  $x$ -axis at the point  $B$ .

- (a) (i) Use the factor theorem to show that  $x - 3$  is a factor of  $p(x) = x^3 - 5x^2 + 7x - 3$ . (2)
- (ii) Hence find the coordinates of  $B$ . (1)
- (b) The point  $M$ , shown on the diagram, is a minimum point of the curve with equation  $y = x^3 - 5x^2 + 7x - 3$ .
- (i) Find  $\frac{dy}{dx}$ . (2)
- (ii) Hence determine the  $x$ -coordinate of  $M$ .

(3)

- (c) Find the value of  $\frac{d^2 y}{dx^2}$  when  $x = 1$ .

(2)

- (d) (i) Find  $\int (x^3 - 5x^2 + 7x - 3) dx$ .

(4)

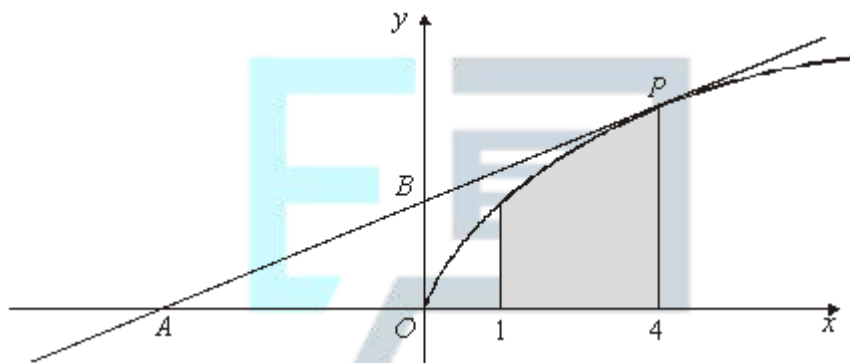
- (ii) Hence determine the area of the shaded region bounded by the curve and the coordinate axes.

(4)

(Total 18 marks)

### Q26.

The diagram shows a curve  $C$  with equation  $y = \sqrt{x}$ . The point  $O$  is the origin  $(0, 0)$ .



The region bounded by the curve  $C$ , the  $x$ -axis and the vertical lines  $x = 1$  and  $x = 4$  is shown shaded in the diagram.

- (a) (i) Write  $\sqrt{x}$  in the form  $x^p$ , where  $p$  is a constant.

(1)

- (ii) Find  $\int \sqrt{x} dx$ .

(2)

- (iii) Hence find the area of the shaded region.

(3)

- (b) The point on  $C$  for which  $x = 4$  is  $P$ . The tangent to  $C$  at the point  $P$  intersects the  $x$ -axis and the  $y$ -axis at the points  $A$  and  $B$  respectively.

- (i) Find an equation for the tangent to the curve  $C$  at the point  $P$ .

(4)

- (ii) Find the area of the triangle  $AOB$ . (3)
- (c) Describe the single geometrical transformation by which the curve with equation  $y = \sqrt{x-1}$  can be obtained from the curve  $C$ . (2)
- (d) Use the trapezium rule with four ordinates (three strips) to find an approximation for  $\int_1^4 \sqrt{x-1} \, dx$ , giving your answer to three significant figures. (4)
- (Total 19 marks)

**Q27.**

- (a) Find  $\frac{dy}{dx}$  when  $y = x \sin 2x$ . (3)
- (b) (i) Find  $\frac{dy}{dx}$  when  $y = (x^2 - 6)^4$ . (2)
- (ii) Hence, or otherwise, find  $\int x(x^2 - 6)^3 \, dx$ . (3)
- (Total 8 marks)